

Name: Shreyans Tatiya

Roll no: 16010123325

Batch : C3 / DV (Fri)

TITLE: Introduction to Power BI

Objective:

To introduce the fundamentals of Power BI and enable students to create basic interactive visualizations using data.

Expected outcome of Experiment:

CO1	Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats.
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Books/ Journals/ Websites referred:

https://learn.microsoft.com/en-gb/training/powerplatform/power-bi?WT.mc_id=sitertzn_learnlab_guidedlearning-card-powerbi&culture=en-us&country=us

Background Theory:

Power BI is a Microsoft business analytics tool that transforms raw data into interactive, visually rich reports and dashboards, helping users discover insights, track trends, and make data-driven decisions. It connects to numerous data sources, offers user-friendly drag-and-drop features for visualization.

Laboratory Work:

Refer this link:

<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-excel-stunning-report>

Perform these tasks

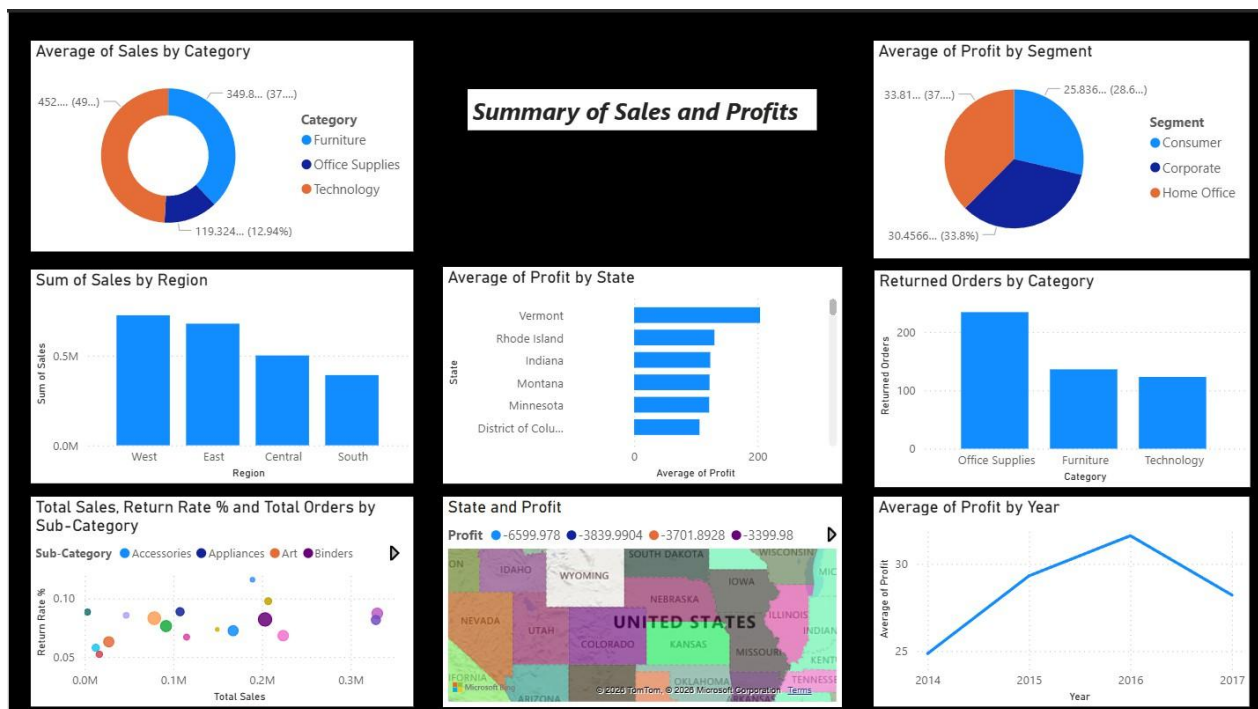
- 1. Connecting to Data Source and Visualizing one specific table**
- 2. Creating and Refining a view**
- 3. Focus results**
- 4. Exploring data geographically**
- 5. Drill down into the details**

Implementation screenshots:

Summary of Sales and Profit for Superstore Dataset

Dashboard includes:

- Avg Sales by Category of Products – Donut Chart
- Avg profit by Segment – Pie Chart
- Sum of Sales by Region – Bar Chart
- Avg Profit by State – Bar Chart
- Returned Orders by Category – Bar Chart
- Total Sales to Return % by Sub Category – Scatter Plot
- Profit by State – Geographic Map
- Avg Profit by year – Line Chart



Conclusion:

The above experiment highlights the major differences between tableau and power bi and gives an introduction to the various features of power bi such as creation of measures, dax, ETL process and a single dashboard with all analysis.

Post Laboratory Questions

- 1. Which tool inside Power BI is used to connect to, clean, and transform data before you start building charts?**

Power Query Editor

It is used to connect to data sources and perform data cleaning, transformation, and preparation before visualization.

- 2. In Power BI, what is the specific language used to write formulas and create custom calculations (like measures and calculated columns)?**

DAX (Data Analysis Expressions)

DAX is used to create measures, calculated columns, and custom calculations in Power BI.

- 3. If you want to show how your Sales have increased or decreased over the last 12 months, which chart type is the standard choice?**

Line Chart

A line chart clearly shows trends and variations in sales over time.

- 4. Your data includes a column for "City" and "Country." Which visual type would automatically plot this data onto a world or regional background?**

Map (Geographical Map Visual)

The map visual automatically plots city and country data on a geographic background.